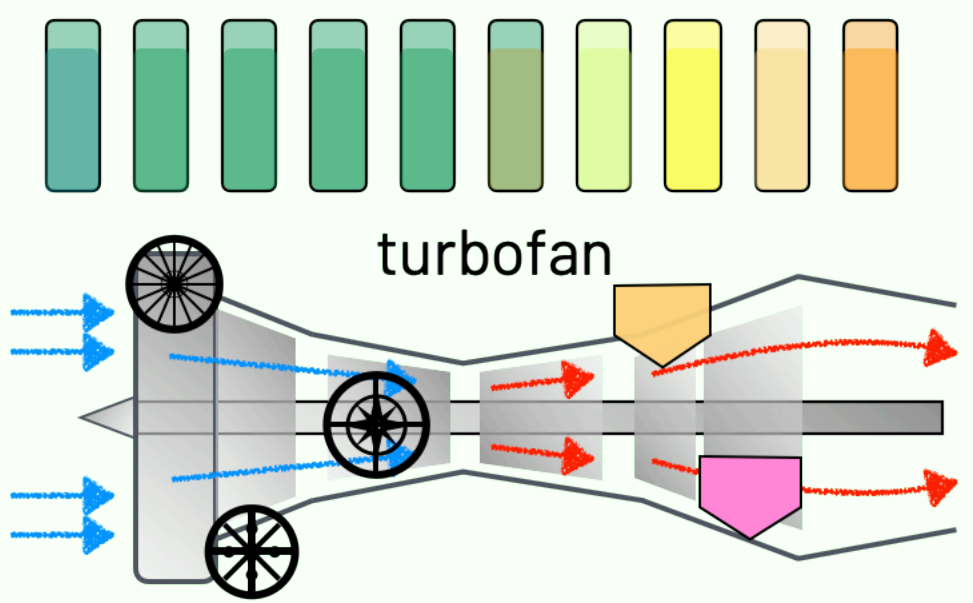




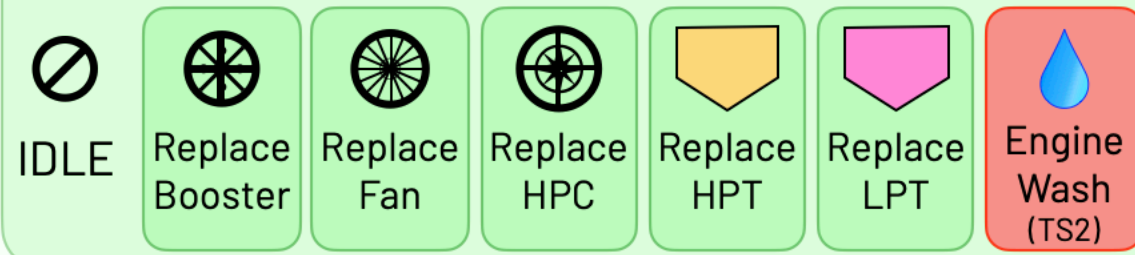
SUBSYSTEM HEALTH AND ACTIONS

Health State $s_t \in \mathbb{R}^{10}$
(efficiency & glow capacity deviations)



Fouling $c_t \in \mathbb{R}_{\geq 0}^5$ (TurboSens2)

MAINTENANCE ACTION a_t

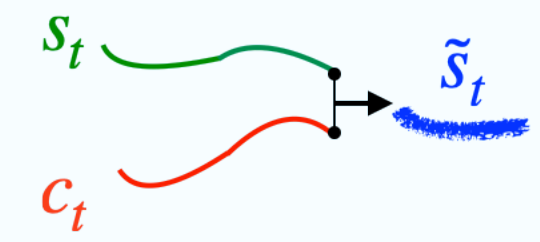


Sparse interventions - Imperfect restoration -
Health-based policy - Opportunistic bundling

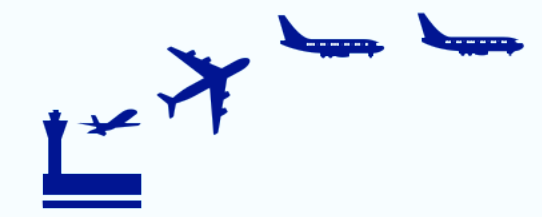


SIMULATOR DYNAMICS OpenDeckSMR (ODSMR)

Combine to
phantom state
 $\tilde{s}_t = f(s_t, c_t)$



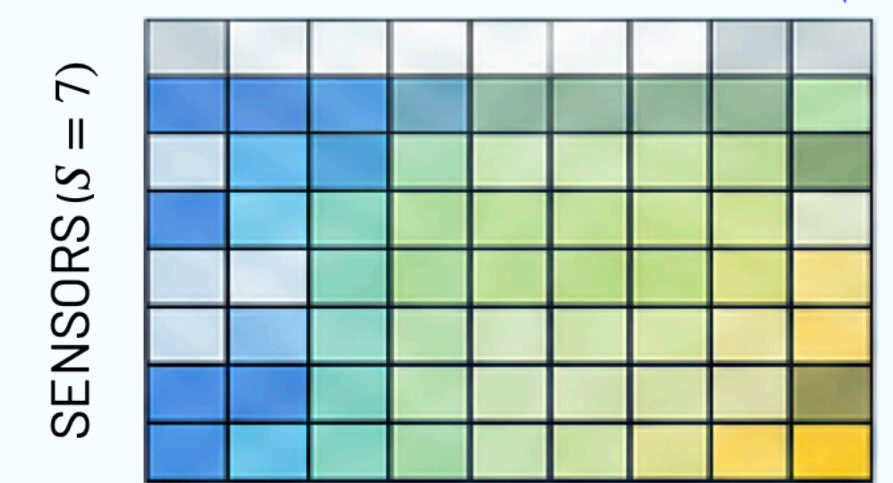
Query across C
flight phase contexts



Sensor observations

$$o_t \in \mathbb{R}^{S \times C}$$

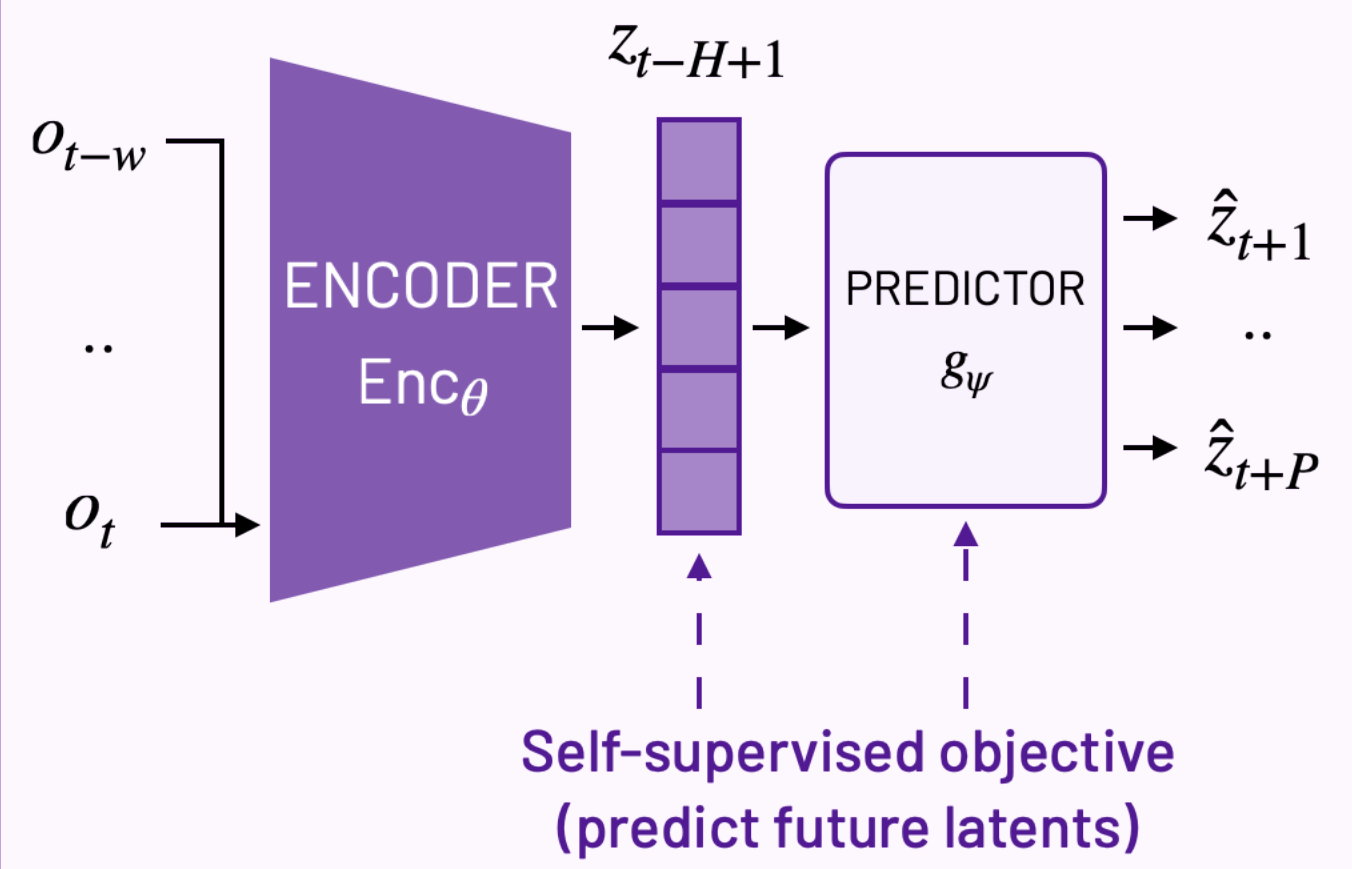
CONTEXTS ($C = 12/16$)



+ SENSOR NOISE



WORLD MODEL REPRESENTATION Self-supervised Pretraining

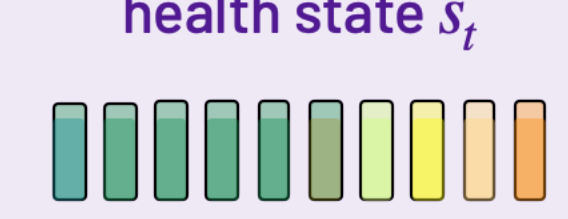


INVERSE PROBING (FROZEN ENCODER)

Fixed linear probe h_ψ

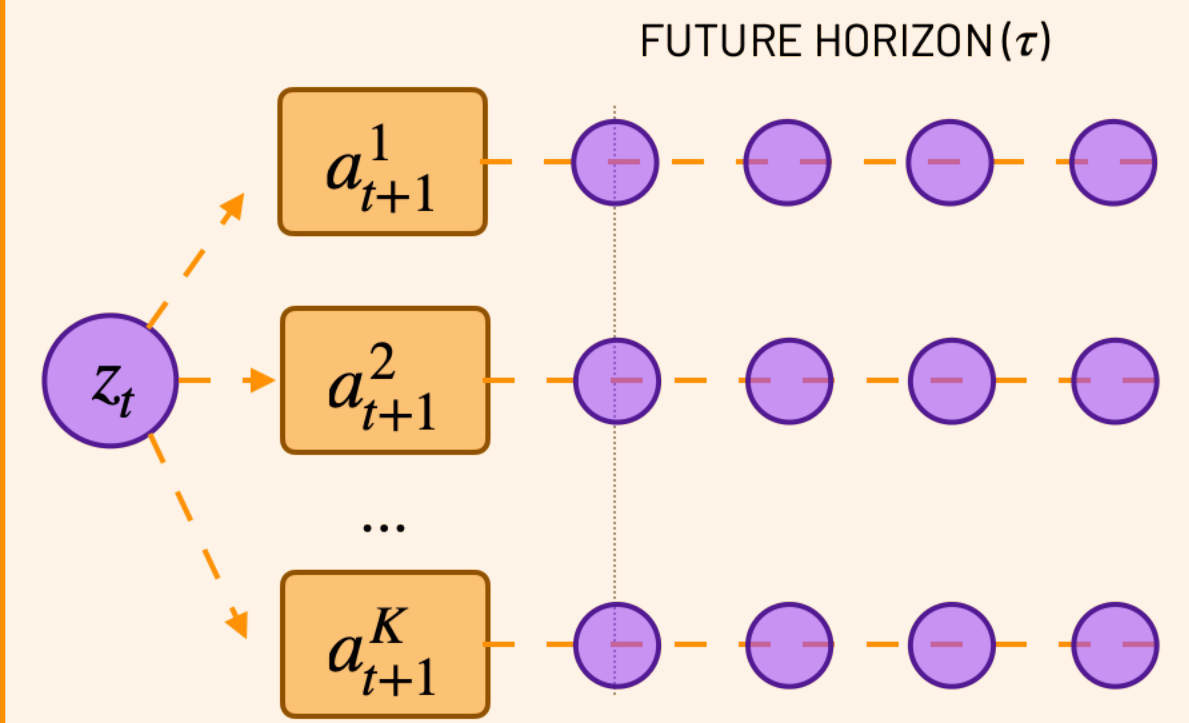


Predict true
health state s_t



COUNTERFACTUAL FORECAST AND EVALUATION

- Counterfactual rollouts under candidate maintenance policies
- Deterministic replay for exact ground-truth comparison
- Evaluate: state recovery, forecasting, OOD generalization



Model prediction



Simulator replay (ground truth)



PAIRED GROUND TRUTH

Sensor stream o_t
paired with full health
state s_t (10D)



INTERACTIVE SIMULATOR

Query online under any
action sequence.
Deterministic replayer
for counterfactuals.



CONFIGURABLE COOKBOOK

Degradation archetypes,
maintenance policies, events,
regions, weather and flight
contexts



TWO SCENARIOS

Turbosens1: 7 x 12 obs.,
6 actions, no fouling
Turbosens2: 7 x 16 obs.,
7 actions, fouling & phantom



REPRODUCIBLE & OPEN

HDF5 datasets, schema docs,
generation scripts, multi-seed
tooling, evaluation pipeline